

Khondokar Radwanur Rahman

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EDUCATION

Rajshahi University of Engineering and Technology

Bachelor of Science in Computer Science

Rajshahi, Bangladesh

Mar 2022 - Jul 2026

PROFESSIONAL EXPERIENCE

Research Assistant

Qatar University

Feb 2025 – Present

Remote

- Enhanced multimodal AI pipelines combining computer vision, time-series sensors, and operational data for industrial anomaly detection and energy forecasting, improving cross-modal prediction reliability by approximately 8%.
- Employed physics-informed neural networks (PINNs) and federated learning to enable data-efficient training across diverse engineering environments handling 100k+ multimodal sensor samples.
- Contributed to 4 Qatar National Research Fund (QNRF) projects by designing scalable machine learning pipelines integrating image, sensor, and structured datasets.
- Co-authored 4 peer-reviewed book chapters submitted to Springer covering multimodal learning, physics informed models, federated learning, and LLM-assisted engineering analytics.

Software Engineer

BengalSub

May 2025 – Jul 2026

Hybrid

- Led the software development of the Autonomous Underwater Vehicles (AUVs) **Hangor 1.0** and **Hangor 2.0** for the international **RoboSub** competition organized by **RoboNation**, California, USA.
- Designed and deployed the autonomy stack on a heterogeneous computing architecture using **NVIDIA Jetson Orin Nano** and **Raspberry Pi 5**, integrating **YOLO**-based object detection, **Depth Anything V2** for monocular depth estimation, and vision-based **pose estimation** for robust underwater perception.
- Developed the mission planning and decision-making framework using **Behavior Trees** with **PyTrees**, enabling modular, fault-tolerant, and autonomous task execution across multiple competition missions.
- Optimized deep learning inference using **TensorRT**, reducing end-to-end perception latency by **68%** and enabling real-time autonomous navigation under resource-constrained embedded hardware.
- Integrated the **Blue Robotics Bar30** pressure sensor and **VectorNav VN-200** IMU for localization, depth estimation, navigation, and closed-loop depth stabilization during autonomous underwater operations.

PROJECTS

- **Tracer:** Developed and deployed a **Multi-Task Physics-Informed Neural Network (MTPINN)** at **HBKU, Qatar** for real-time flow regime classification and gas/liquid velocity estimation, integrating momentum-balance constraints and a multimodal sensor-to-vision retrieval framework for digital twin validation, achieving **96.12% $\pm$ 2.3%** classification accuracy.
- **AgroVision:** Built a scalable plant disease classification system using **TensorFlow** with an end-to-end MLOps pipeline, integrating **DVC** for data/model versioning, **MLflow** for experiment tracking, Dockerized deployment on **AWS** with GitHub Actions CI/CD, and real-time inference through a **Flask** web application.
- **BijoyTori1.0:** Architected a **ROS2 Jazzy** autonomy pipeline on **Raspberry Pi**, integrating RGB-D perception, LiDAR-based obstacle detection, and **EKF** sensor fusion with **Intel RealSense D455** and **RPLidar A2**; implemented autonomous navigation using global/local planners and dynamic costmaps, and validated the system in **Gazebo Classic** prior to hardware deployment.

RESEARCH AND PUBLICATIONS

- **Khondokar Radwanur Rahman**, Amith Khandakar, Md. Fahim Hossen, Shaikh Golam Rabbani, Anirudha Saha, Md. Faysal Ahamed, Md. Munawar Hossain, Mohamed Arselene Ayari, Rajab Abdulla R.E. Al-Esmail, ReliefNet: A Knowledge-Driven, Explainable AI Multimodal Framework for Disaster Severity Classification and Humanitarian Decision-Making, Progress in Disaster Science (**Q1**), Volume 29, 2026, 100528, ISSN 2590-0617.

- **Khondokar Radwanur Rahman**, Amith Khandakar, Abinash Barooah, Mohamed Arselene Ayari, Mohammad Azizur Rahman, Multiphase system leak detection, localization, and dimension estimation using multi-head Physics-informed neural network, Process Safety and Environmental Protection (**Q1**), Volume 211, 2026, 108532, ISSN 0957-5820.
- Amith Khandakar, David G. Michelson, Shaikh Golam Rabbani, Fariya Bintay Shafi, Md. Faysal Ahamed, **Khondokar Radwanur Rahman**, Md Abidur Rahman, Md. Fahmidun Nabi, Mohamed Arselene Ayari, Khaled Khan, Ponnuthurai Nagaratnam Suganthan, RoadSens-4M: A Multimodal Smartphone & Camera Dataset for Holistic Road-way Analysis, Sci Data (2026). <https://doi.org/10.1038/s41597-026-07072-y>. (**Q1**)
- Rahul Debnath, Md. Tajim An Noor, **Khondokar Radwanur Rahman**, Md. Monuwar Hussien, Md. Faysal Ahamed, Moloy Kumar Ghosh, ROS2-Based Autonomous Wheelchair with Adaptive Navigation and LLM-Driven Conversational Control, 28th International Conference on Computer and Information Technology, 2025, <https://doi.org/10.1109/ICCIT68739.2025.11490350>.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL

Machine Learning: PyTorch, TensorFlow, Scikit-Learn, OpenCV, YOLO

Robotics & Autonomy ROS2, MavLink, Pixhawk, Behavior Tree, Gazebo.

Sensors & Perceptions: LiDAR, RGB-D Cameras, Kalman Filters, PointCloud Processing.

MLOps & Infrastructure: AWS, Docker, DVC, MLflow, Git.

Web Technologies: FastAPI, Flask, Django, React, React Native.

HONORS & ACHIEVEMENTS

- **Semifinalist** in RoboSub 2026, and **18th/60** teams in Design Documentation in California, USA.
- Represented Bangladesh in **RoboBoat 2026** competition in Florida, USA.
- Led **RUET Exover** in Mars Autonomous Rover Rally 2026 and ranked **6th** globally held by NASA Texas Space Grant.
- Represented Bangladesh in **RoboSub 2025** competition in California, USA.
- Secured **15th/355** teams globally in Global AI Hackathon held by Kaggle.
- Secured **12th/103** teams globally in Synthetic 2 Real Object Detection Challenge by Kaggle.
- Maximum Rating **Pupil, 1216** in Codeforces.
- Ranked **7th/135 teams** in 5th International KIBO-RPC national round.
- **Finalist** in National Hackathon at East West University.
- Our team **MostQuitt** secured **2nd** place and funded pre-seed of 60,000.
- Received technical scholarship 3 times (2022, 2023, 2024) from RUET.
- Titled accomplished competitor in University Physics Olympiad 2023.
- Our team **Best_Trio** secured **3rd** place among 100+ teams nationally.

VOLUNTEERING EXPERIENCE

- Research Lead, ECE Club, RUET.
- Class Representative, ECE 20 Series, RUET.
- Executive Member, BASIS Student Chapter, RUET.

RECOMMENDATIONS

Md. Faysal Ahamed

Assistant Professor, Department of Electrical & Computer Engineering

Rajshahi University of Engineering and Technology (RUET)

Email: faysal.ahamed@ece.ruet.ac.bd

Google Scholar: Md Faysal Ahamed *

Amith Khandakar Abdullah

Senior Lecturer, Dept. of Electrical Engineering

Top 2% scientist globally according to Elsevier Report 2023

Email: amithk@qu.edu.qa

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